

Dimension reduction

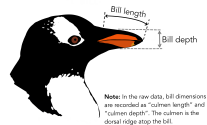
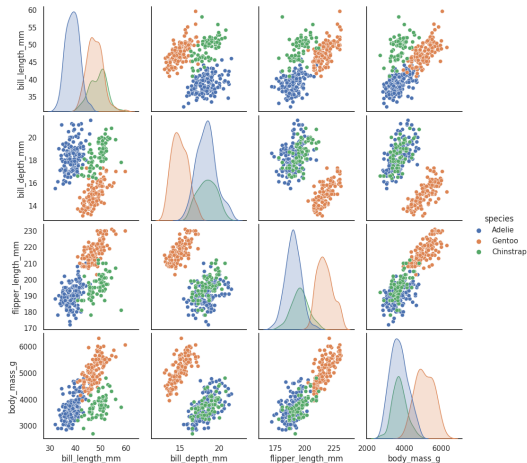
Master MIASHS

Paul Tresson

18/10/25

Introduction

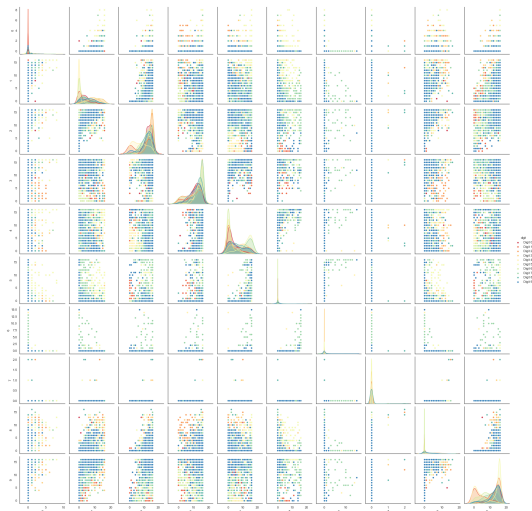
Visualize what is happening in higher dimensions



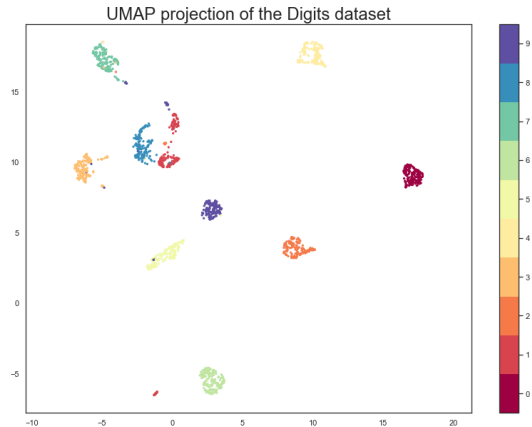
Visualize what is happening in higher dimensions

0 0 8 3 1 4 5 0 0 2 1 9 7 4 4 7 4 2 6 4
1 1 4 7 7 4 0 1 2 2 5 4 7 5 3 6 1 9 0
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4 4 7 4 2 6 4 4 6 2 5 8 2 8 8 0 4 7 9
5 5 3 6 1 3 9 5 0 4 8 7 9 9 0 9 3 5 3
6 6 5 6 7 6 0 6 0 1 2 3 5 0 2 1 4 4 0
7 7 1 6 4 1 7 7 3 2 1 4 4 3 2 5 3 4 1
8 8 0 4 6 7 9 8 6 0 4 8 2 5 7 0 9 7 2
9 9 0 9 3 5 8 3 3 0 0 8 3 5 8 9 1 2 3
0 0 2 1 1 4 0 2 3 1 7 7 4 5 6 2 5 7 8 4
1 1 2 5 3 9 7 2 2 4 3 6 6 0 6 4 8 8 2 6
2 5 9 0 3 1 2 3 7 3 3 3 9 8 7 9 2 2 4 5 7
3 5 8 3 1 2 3 7 3 3 3 3 8 7 9 2 2 4 5 7
4 6 2 5 7 8 4 4 7 4 2 6 9 8 8 6 0 3 7 8
5 5 0 2 6 2 5 5 3 6 4 4 8 9 9 3 0 1 9 9
6 0 1 8 8 2 6 6 5 6 7 4 7 0 8 3 1 4 5 0
7 5 2 2 4 5 7 7 1 6 3 5 4 4 4 7 7 0 4 1
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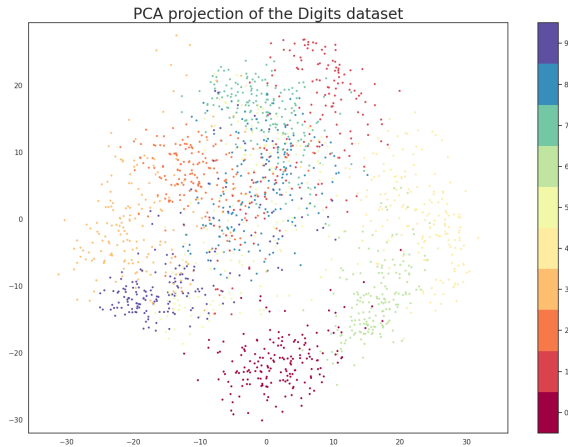
Visualize what is happening in higher dimensions



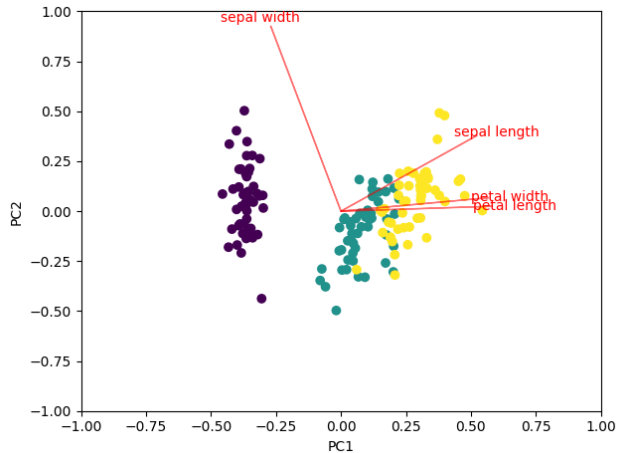
Visualize what is happening in higher dimensions



PCA



PCA interpretability



PCA advantages and drawbacks

Advantages

- fast

Drawbacks

PCA advantages and drawbacks

Advantages

- fast
- scales well

Drawbacks

PCA advantages and drawbacks

Advantages

- fast
- scales well
- explanatory

Drawbacks

PCA advantages and drawbacks

Advantages

- fast
- scales well
- explanatory

Drawbacks

- not suited for non-linear data

Non linear data

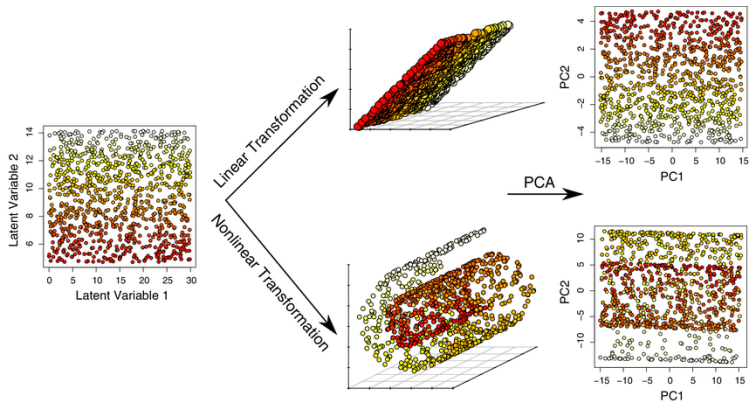


Figure from Du, 2019

Introduction

UMAP

How to work with non linear data

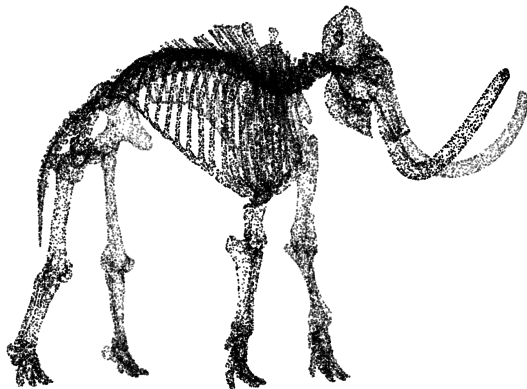
1. Find a good representation of the data in high dimension
2. Fit a good representation of in low dimension

- **SNE** (Stochastic Neighbor Embedding) Hinton and Roweis, 2002

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- **T-SNE** (T-distributed SNE) Van der Maaten and Hinton, 2008

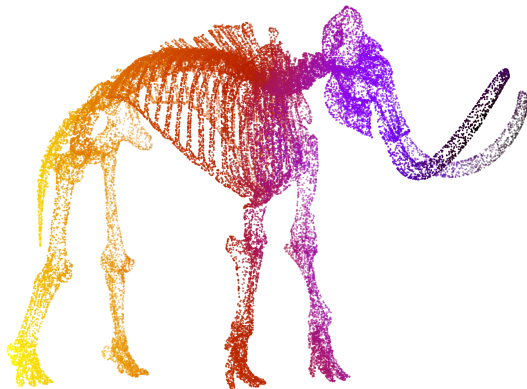
- **SNE** (Stochastic Neighbor Embedding) Hinton and Roweis, 2002
- **T-SNE** (T-distributed SNE) Van der Maaten and Hinton, 2008
- **UMAP** (Uniform Manifold Approximation and Projection) McInnes et al., 2018

Mammoth example



Mammoth dataset by the Smithsonian

Mammoth example



Mammoth dataset by the Smithsonian

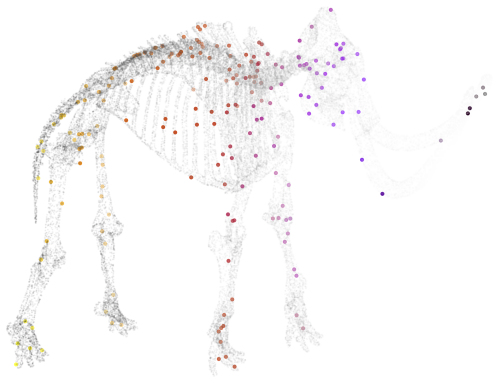
Mammoth example - PCA



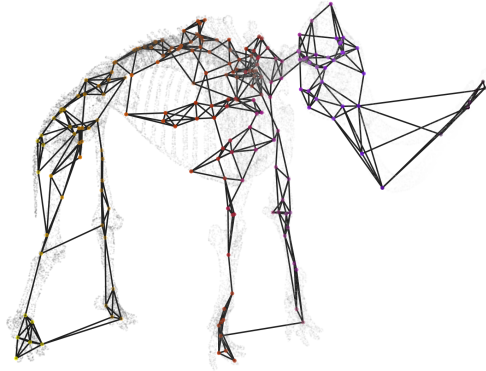
Finding high dimension graph



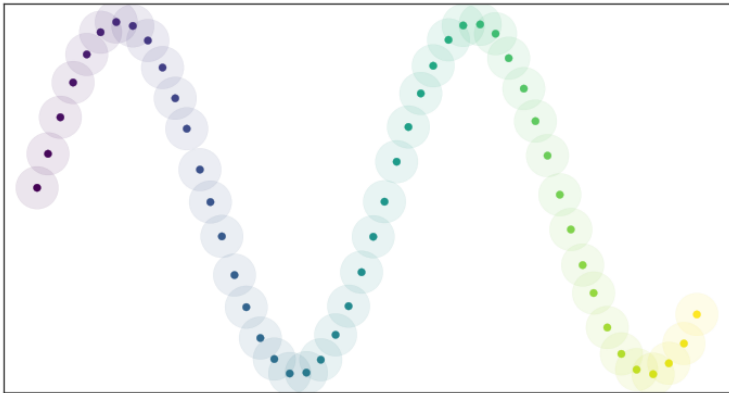
Finding high dimension graph



Finding high dimension graph

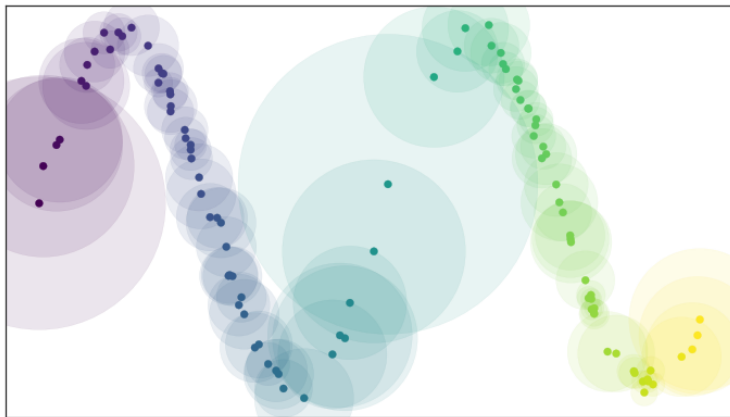


Uniform Manifold Approximation and Projection



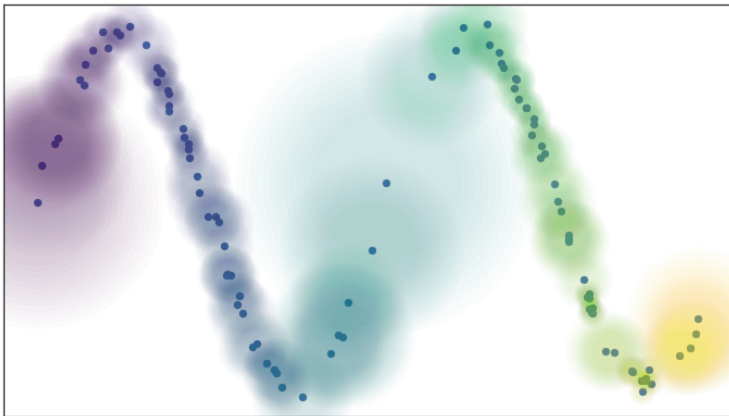
[UMAP documentation](#)

Uniform Manifold Approximation and Projection



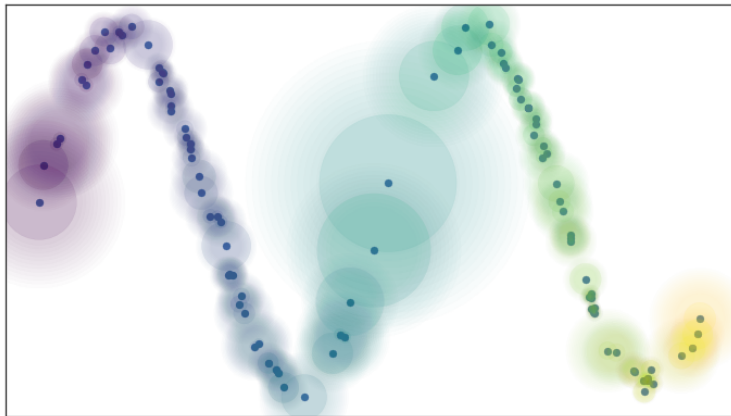
UMAP documentation

Uniform Manifold Approximation and Projection



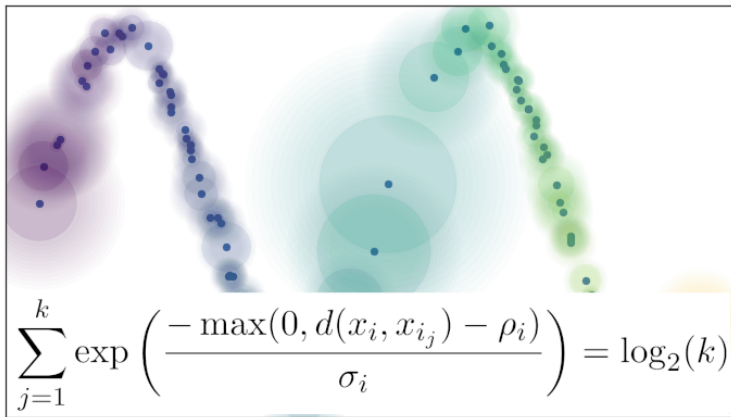
[UMAP documentation](#)

Uniform Manifold Approximation and Projection



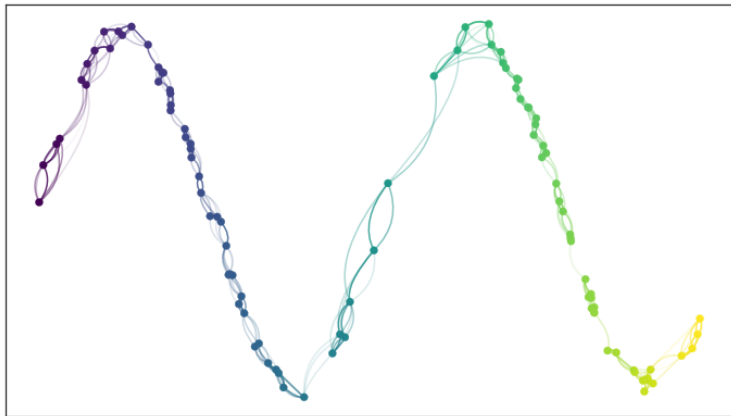
[UMAP documentation](#)

Uniform Manifold Approximation and Projection



[UMAP documentation](#)

Uniform Manifold Approximation and Projection



[UMAP documentation](#)

Uniform Manifold Approximation and Projection

Definition 9. Let $X = \{X_1, \dots, X_N\}$ be a dataset in $\mathbb{R}^n$. Let $\{(X, d_i)\}_{i=1 \dots N}$ be a family of extended-pseudo-metric spaces with common carrier set X such that

$$d_i(X_j, X_k) = \begin{cases} d_{\mathcal{M}}(X_j, X_k) - \rho & \text{if } i = j \text{ or } i = k, \\ \infty & \text{otherwise,} \end{cases}$$

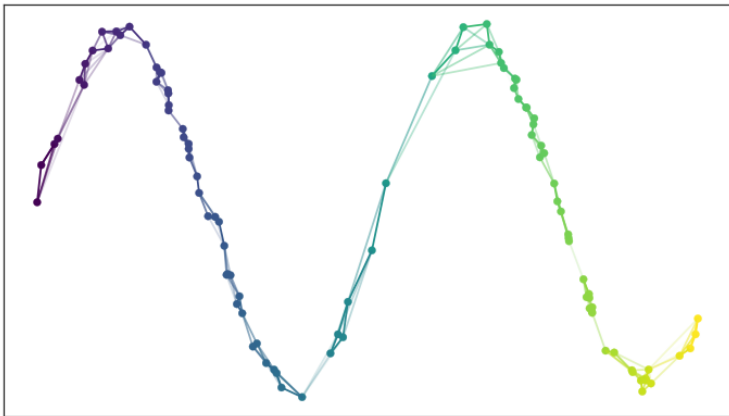
where ρ is the distance to the nearest neighbor of X_i and $d_{\mathcal{M}}$ is geodesic distance on the manifold $\mathcal{M}$, either known apriori, or approximated as per Lemma 1.

The fuzzy topological representation of X is

$$\bigcup_{i=1}^n \text{FinSing}((X, d_i)).$$

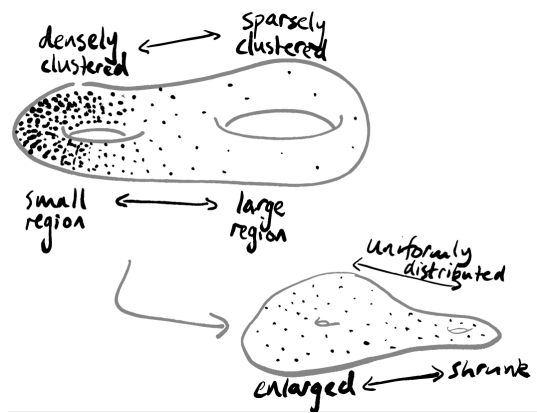
Fuzzy *Topology*

Uniform Manifold Approximation and Projection

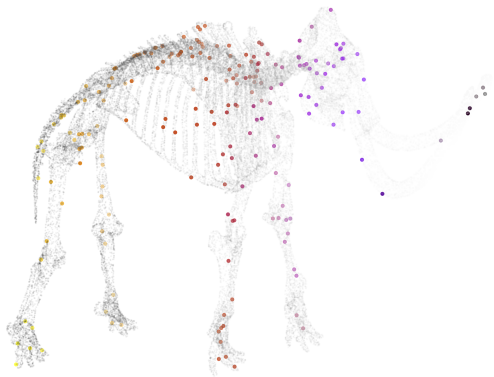


UMAP documentation

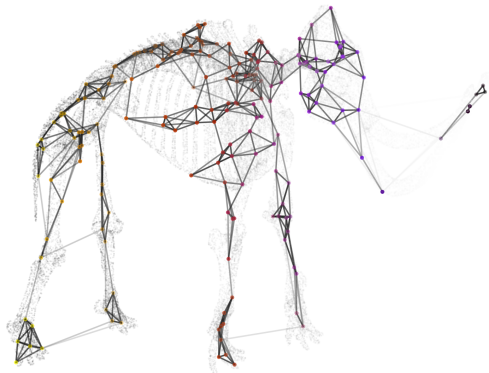
Uniform Manifold Approximation and Projection



Finding high dimension graph



Finding high dimension graph



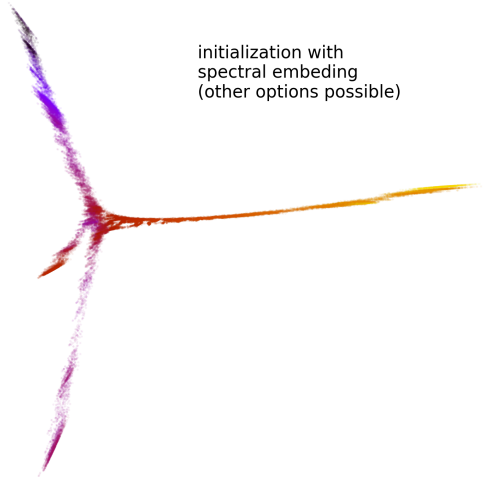
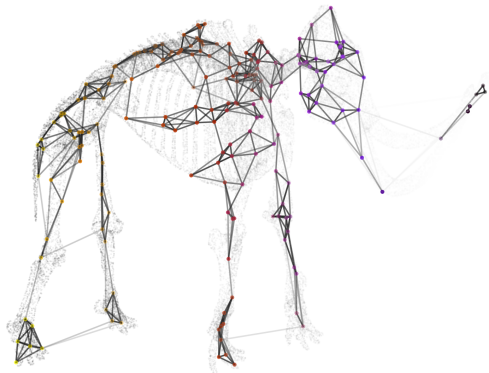
Fitting low dimensional representation



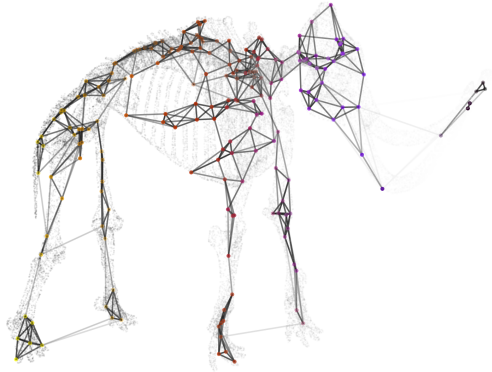
Fitting low dimensional representation



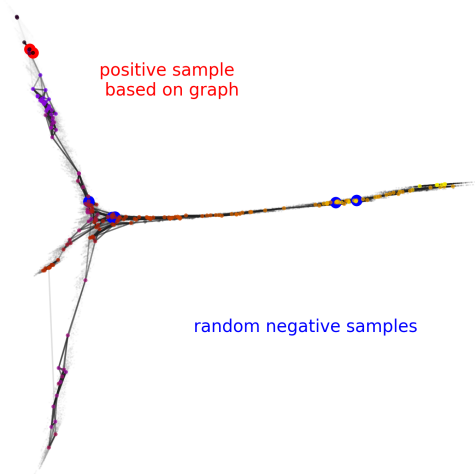
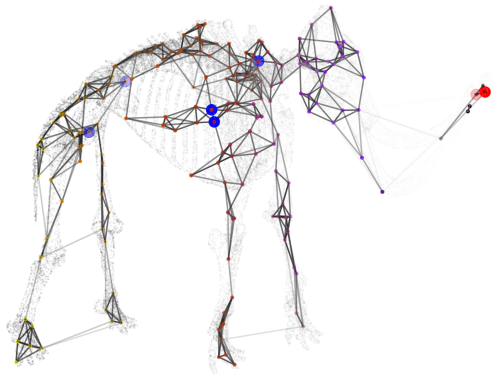
Fitting low dimensional representation



Fitting low dimensional representation



Fitting low dimensional representation



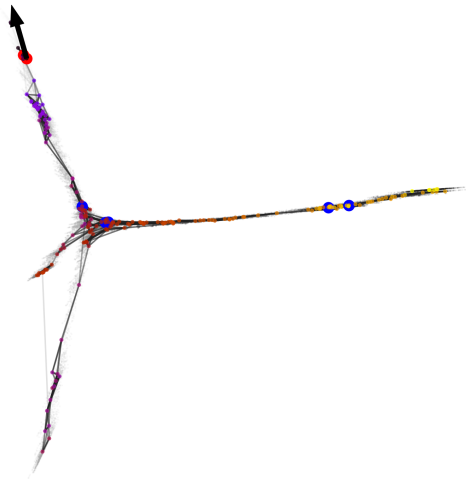
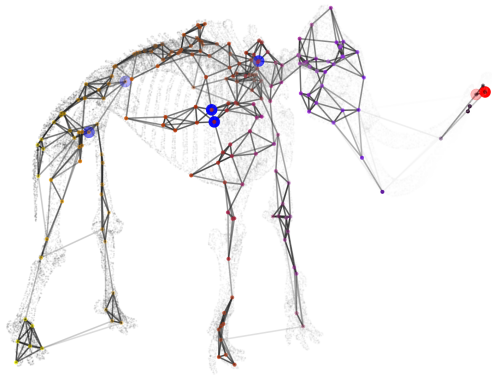
Fitting low dimensional representation

Cross-entropy loss counting **positive** and **negative** attraction forces

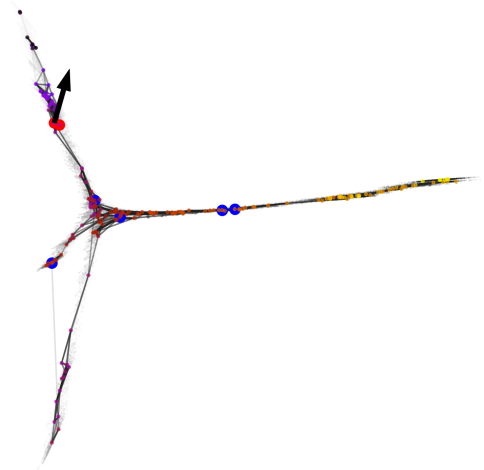
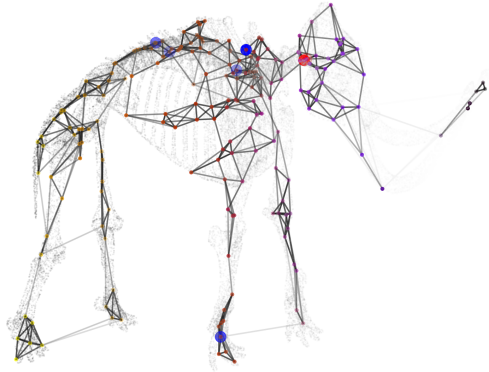
$$C_{UMAP} = \sum_{i \neq j} v_{ij} \log \left(\frac{v_{ij}}{w_{ij}} \right) + (1 - v_{ij}) \log \left(\frac{1 - v_{ij}}{1 - w_{ij}} \right) \quad (1)$$

Optimization with **Stochastic Gradient Descent**

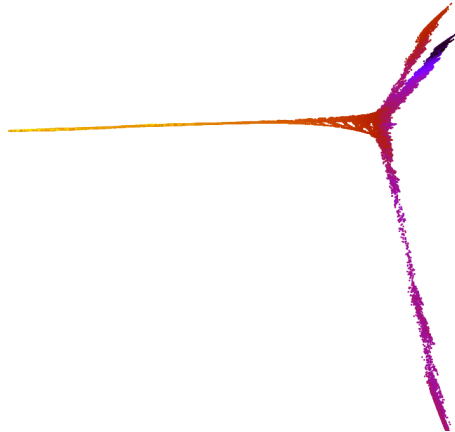
Fitting low dimensional representation



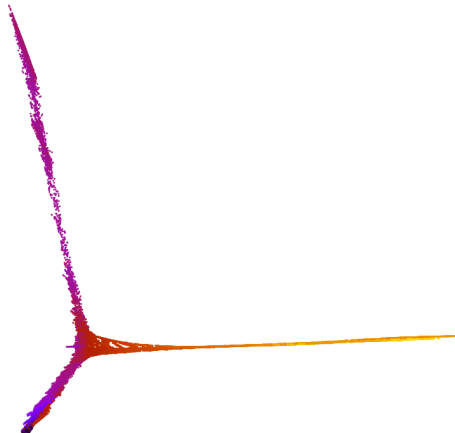
Fitting low dimensional representation



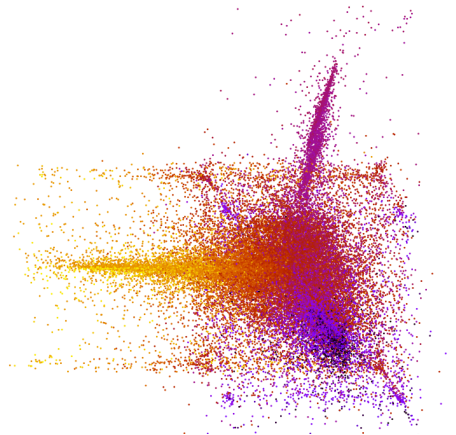
Optimization



Optimization



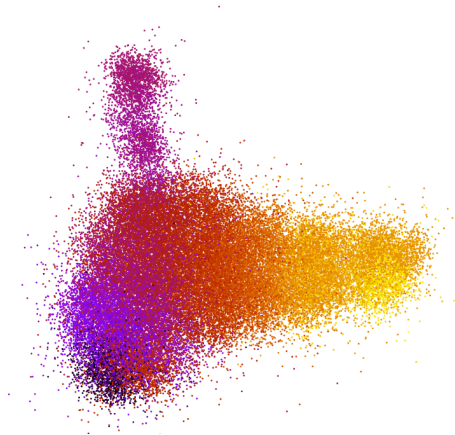
Optimization



Optimization



Optimization



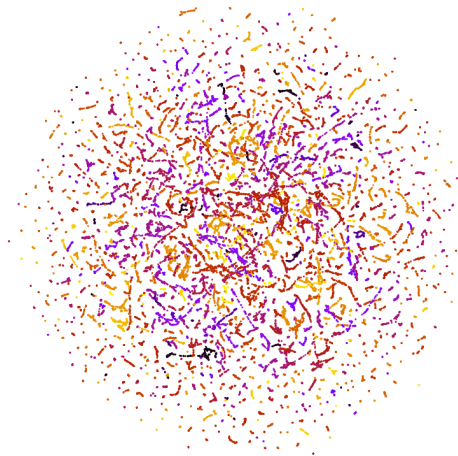
Optimization



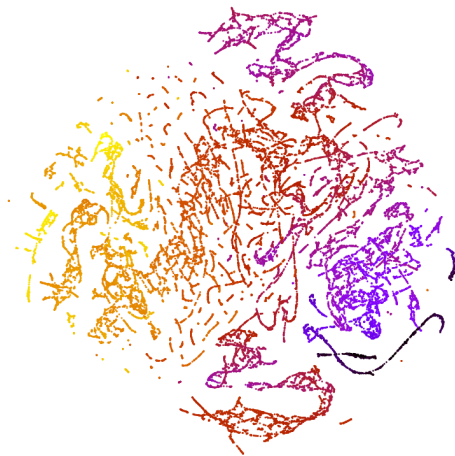
Hyper-parameters

- number of neighbors
- min distance
- distance metric

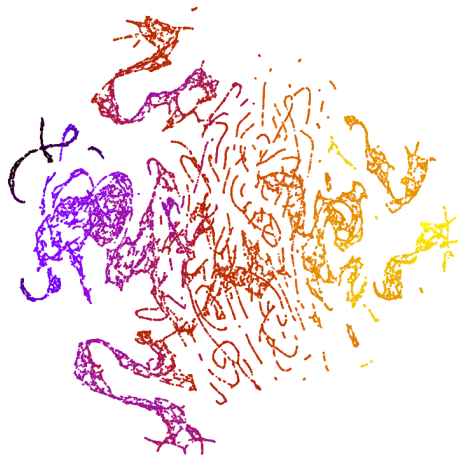
Influence of the number of neighbors



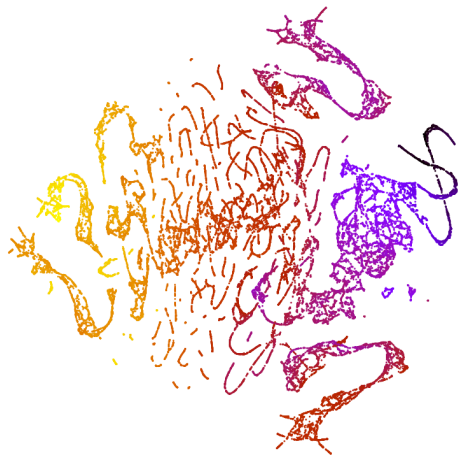
Influence of the number of neighbors



Influence of the number of neighbors



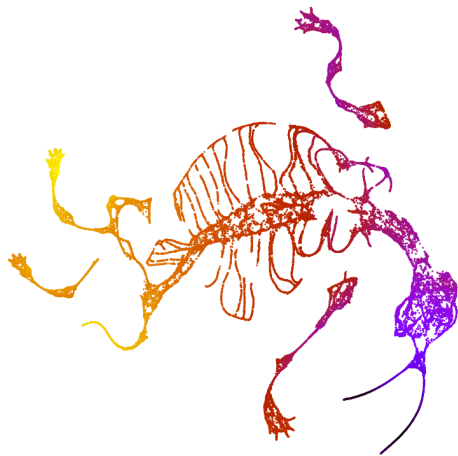
Influence of the number of neighbors



Influence of the number of neighbors



Influence of the number of neighbors



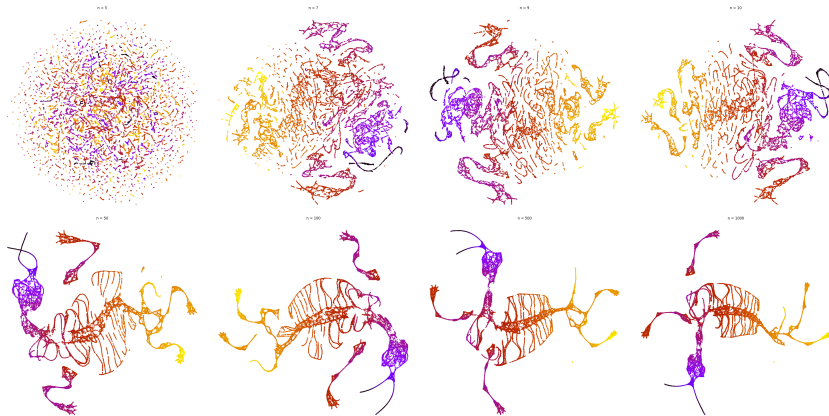
Influence of the number of neighbors



Influence of the number of neighbors



Influence of the number of neighbors



UMAP vs. T-SNE

UMAP

- graph theory, fuzzy logic

T-SNE

- probabilities

UMAP vs. T-SNE

UMAP

- graph theory, fuzzy logic
- determinist initialization

T-SNE

- probabilities
- random initialization

UMAP vs. T-SNE

UMAP

- graph theory, fuzzy logic
- determinist initialization
- $\log_2$ similarity scores

T-SNE

- probabilities
- random initialization
- gaussian distribution similarity

UMAP vs. T-SNE

UMAP

- graph theory, fuzzy logic
 - determinist initialization
 - $\log_2$ similarity scores
 - update by pairs
- scales well for large datasets

T-SNE

- probabilities
 - random initialization
 - gaussian distribution similarity
 - update all points
- scales less

Scaling

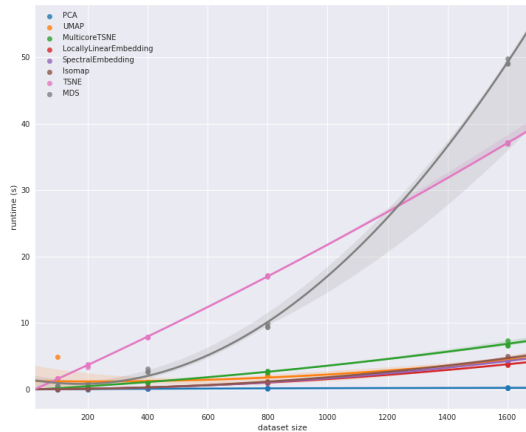
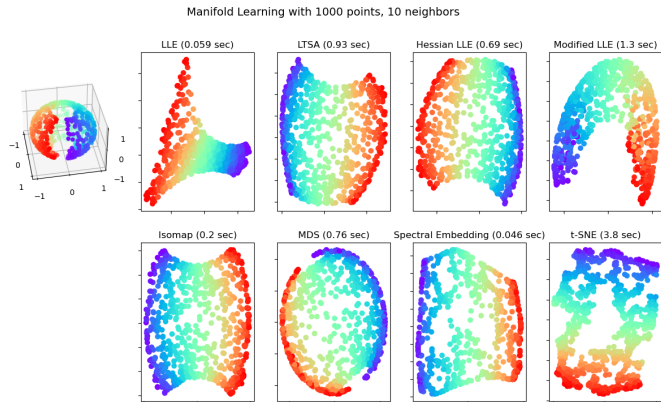


Figure from umap-learn documentation

A lot of options !



Useful ressources

- `scikit-learn docs` !
- `deepia`
- `StatQuest`

Thanks for your attention !

Let's practice !

References i

- Du, Trina Y (2019). “**Dimensionality reduction techniques for visualizing morphometric data: Comparing principal component analysis to nonlinear methods**”. In: *Evolutionary Biology* 46.1, pp. 106–121.
- Hinton, Geoffrey E and Sam Roweis (2002). “**Stochastic neighbor embedding**”. In: *Advances in neural information processing systems* 15.
- McInnes, Leland, John Healy, and James Melville (2018). “**Umap: Uniform manifold approximation and projection for dimension reduction**”. In: *arXiv preprint arXiv:1802.03426*.
- Van der Maaten, Laurens and Geoffrey Hinton (2008). “**Visualizing data using t-SNE.**”. In: *Journal of machine learning research* 9.11.